

Abelardo Diaz Sanchez

Full-Stack Developer & AI Agent Architect

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PROFILE

Full-stack developer specialized in AI agents built on Claude. I design, build, and operate production systems: multichannel conversational agents (WhatsApp, Telegram, web) with RAG and guardrails, multi-tenant SaaS, synced ecommerce, and integrations with real hardware (CCTV, GPS). Everything runs on infrastructure I manage myself: Linux VPS, PostgreSQL, Docker, PM2, Cloudflare. Before code: 10 years founding and operating REDV6, a WISP serving rural communities in San Luis Potosi, Mexico, 24/7. Verifiable portfolio with case studies: abelardodiaz.dev.

SELECTED PROJECTS

Agente 080

2026

Multichannel AI agent (WhatsApp, Telegram, web) that goes beyond replying: it reasons, decides, and executes real business actions — collections, sales, support, scheduling, real-estate inventory — with multiple specialist agents, memory, RAG, and guardrails. Bounded autonomy: every action has defined limits and is auditable.

Python FastAPI Celery Redis PostgreSQL Claude API

Internal platform: Debates IA + Manager internal use, in production

2025-2026

Two APIs that govern my operation: (1) a unified gateway over 8 AI providers (OpenAI, Gemini, DeepSeek, Grok, Groq, Qwen, Kimi + web search) with persistent conversations, multi-AI debates with automatic consensus, a knowledge base that intercepts queries, per-project auth with key rotation, and an MCP server consumable from Claude Code; (2) a centralized project manager: fleet-wide catalog and health checks, git progress tracking, Cloudflare management (DNS/SSL/cache) with granular per-key permissions, and RAG on pgvector (hybrid vector + full-text search with RRF fusion).

Python PostgreSQL + pgvector MCP REST APIs Cloudflare API

citas-bot-universal open source, MIT

2026

Python template for WhatsApp appointment assistants built on Claude: FastAPI + SQLite + APScheduler. Distilled from a private production system for medical practices; ready to fork and deploy.

Python FastAPI Anthropic API WhatsApp Business API

Video Frame Analyzer

2026

CLI that extracts video frames with FFmpeg scene detection, analyzes them with Gemini Flash (vision), and generates a narrative with Claude. Multi-model pipeline at ~\$0.008 USD per video.

Python FFmpeg Gemini Vision Claude Pydantic

Domus Rentas: multi-tenant SaaS

2025-2026

Evolution of a production rental platform into a multi-tenant SaaS built as 9 polyglot microservices: Django where it fits, FastAPI where real time matters, and a Next.js frontend acting as gateway.

Django FastAPI Next.js PostgreSQL Redis Docker

Tienda Django Sync

2026

Ecommerce whose real engine is not the storefront: automatic sync of catalog, prices, inventory, and exchange rate against the Syscom wholesaler API.

Django Next.js Celery PostgreSQL Docker

ulfbk - Recyclable Blocks open source

2026

Ecosystem of 28 composable multi-stack packages (Python + TypeScript): reusable infrastructure for building APIs, multitenant SaaS, bots, and dashboards. Monorepo with uv + pnpm + Turborepo, zero circular dependencies.

Python TypeScript FastAPI Next.js Turborepo

Call Blocker open source, published on F-Droid

2026

Android app for blocking unwanted calls, published on F-Droid (passes the store's reproducibility and licensing review). Kotlin + Jetpack Compose + Clean Architecture.

Kotlin Jetpack Compose Material 3 Hilt Room

EXPERIENCE

Full-Stack Developer & AI Agent Architect — Independent

2024 – present

I build and operate production systems for clients in Mexico and Europe with an AI-native workflow: spec-driven development with Claude Code as my primary tool, reusable skills and commands portable across projects, persistent memory, MCP integrations (own servers and clients), and orchestration of 8 AI models by cost and context. Full cycle: architecture, implementation, deployment, and operations on my own servers, with a secure publication pipeline (gitleaks + custom sanitization). Beyond the selected projects: video surveillance backends (Hikvision ISAPI, 19 REST endpoints) and GPS fleet tracking (Tracksolid, 26 endpoints), verified against real hardware from my own operation.

Founder & Technical Director — REDV6 (WISP)

2015 – present

Wireless ISP bringing connectivity to rural communities in San Luis Potosi underserved by major carriers. 24/7 operations with residential and SMB customers.

- Design and operation of long-distance point-to-point links at 2.4/5/60 GHz (Cambium, MikroTik, Altai).
- Enterprise networks, high-coverage WiFi, IP PBX systems, CCTV, and GPS fleet tracking.
- Off-grid solar backup power systems for remote sites.
- Today it serves as the real-hardware lab where I verify my software integrations.

Sales Executive — AT&T Mexico

2013 – 2015

Consultative sales of technology and internet services to businesses.

Sales Executive — Axtel Telecomunicaciones

2010 – 2013

Consultative sales of technology and internet services to businesses.

Digital Marketing & Online Business Consultant — Consultoria e Internet para Empresas (WSI franchise)

2005 – 2010

Digital marketing consulting and online business development for SMBs under the international WSI franchise methodology.

IT & Networking Instructor — Tec. en Sistemas (Salinas de Hidalgo) / Inst. Quetzal (Zacatecas)

2001 – 2003

Taught LAN networking, Windows 2000 Server, and Linux (Red Hat 8, Mandrake 9); PC assembly and maintenance.

SKILLS

Languages	Python, TypeScript, SQL, Kotlin (Android)
AI & agents	Claude API (agents, tool use), Claude Code, MCP (servers and clients), RAG (pgvector, hybrid search), multi-provider orchestration (8 models), spec-driven development, Whisper, Gemini Vision, context engineering
Backend	FastAPI, Django, Flask, Celery, Redis, PostgreSQL, WhatsApp Business API, webhooks (HMAC, retry queues), Stripe
Frontend	Next.js, React, Tailwind CSS, Astro
Infra / DevSecOps	Linux (self-hosted VPS), Docker, PM2, Cloudflare, Vercel, GitHub Actions, gitleaks + pre-publication sanitization, MikroTik/Cambium networking

CERTIFICATIONS

MTCNA - Network Associate	MikroTik, 2018	MTCRE - Routing Engineer	MikroTik, 2018
ePMP - Wireless links	Cambium Networks, 2018	Super-coverage WiFi	Altai Technologies, 2018

EDUCATION & LANGUAGES

Computer Systems Engineering — Instituto Tecnológico de Zacatecas (partial coursework)	2000 – 2004
Spanish: native · English: technical — fluent reading, working conversation.	